

Variety $curve_q4$ over GF(4)

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The equation

The equation of the variety is :

$$X_0^4 + X_1^4 + X_2^4 + X_0^2 X_1^2 + X_0^2 X_2^2 + X_1^2 X_2^2 + X_0^2 X_1 X_2 + X_0 X_1^2 X_2 + X_0 X_1 X_2^2 = 0$$

(1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1)

$$X_0^4 + X_1^4 + X_2^4 + X_0^2 X_1^2 + X_0^2 X_2^2 + X_1^2 X_2^2 + X_0^2 X_1 X_2 + X_0 X_1^2 X_2 + X_0 X_1 X_2^2 = 0$$

Summary

Number of automorphisms	−1
Order of set stabilizer	336
Number of points	14
Number of lines	−1
Number of singular points	−1

Set stabilizer

The stabilizer of order 336 is generated by:

$$g_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_1$$

$$g_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_0$$

$$g_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}_1$$

$$g_4 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_1$$

$$g_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}_0$$

$$g_6 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}_1$$

1,0,0,0,1,0,0,0,1,1
 1,0,0,0,1,0,0,1,1,0
 1,0,0,0,1,0,1,0,1,1
 1,0,0,1,1,0,0,0,1,1
 1,0,0,0,0,1,0,1,0,0
 0,1,0,1,0,0,0,1,1,1

The variety has 14 points. They are: (5, 6, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20)

0 : $P_5 = (2, 1, 0)$	5 : $P_{12} = (3, 1, 1)$	10 : $P_{17} = (0, 3, 1)$
1 : $P_6 = (3, 1, 0)$	6 : $P_{13} = (0, 2, 1)$	11 : $P_{18} = (1, 3, 1)$
2 : $P_8 = (2, 0, 1)$	7 : $P_{14} = (1, 2, 1)$	12 : $P_{19} = (2, 3, 1)$
3 : $P_9 = (3, 0, 1)$	8 : $P_{15} = (2, 2, 1)$	13 : $P_{20} = (3, 3, 1)$
4 : $P_{11} = (2, 1, 1)$	9 : $P_{16} = (3, 2, 1)$	